Co-Evolutionary Feature Compression for Early-Stage Chronic Kidney Disease Detection on MIMIC-IV

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***Abstract*—Early identification of chronic kidney disease (CKD) remains constrained by the cost and logistical complexity of comprehensive renal panels, particularly in primary-care environments and low-resource health systems where laboratory throughput is limited. This work introduces a co-evolutionary optimization pipeline that simultaneously compresses the input feature space and adapts classifier hyperparameters through a shared bipartite fitness matrix. The architecture couples three nature-inspired metaheuristics—the Starfish Optimization Algorithm for binary feature mask navigation, the Dream Optimization Algorithm for continuous hyperparameter tuning, and the Fungal Growth Optimizer for stagnation-breaking perturbation—within a unified optimization framework that eliminates the decoupling bias inherent in sequential pipelines. Evaluation was conducted on a heterogeneous critical-care cohort drawn from the MIMIC-IV database under a strict nested 10×3 cross-validation protocol, with synthetic minority oversampling restricted to inner training partitions to prevent leakage-induced inflation of validation metrics. The selected feature subset substitutes administrative and demographic context for redundant biochemical assays, suggesting a methodologically rigorous pathway toward affordable population-scale screening while preserving discriminative performance. The findings indicate that a lightweight, interpretable feature-compression strategy may complement temporal deep-learning approaches in operational clinical workflows.**

***Keywords*—Chronic Kidney Disease, Feature Selection, Co-Evolutionary Optimization, MIMIC-IV, Clinical Machine Learning, Metaheuristics.**

I. INTRODUCTION

Chronic kidney disease (CKD) constitutes one of the most consequential non-communicable diseases of the present era, with global prevalence estimates approaching 850 million individuals and a disability-adjusted life-year burden exceeding several malignancies of comparable mortality [1], [2]. The clinical trajectory of CKD is characterized by progressive nephron loss, frequently silent until estimated glomerular filtration rate descends below 60 mL/min/1.73m², at which stage therapeutic windows for nephroprotection narrow substantially. Diagnostic guidelines codified by KDIGO 2024 [3] mandate confirmation through serum creatinine, urinary albumin-to-creatinine ratio, and complementary biochemical panels, the cumulative cost of which renders routine population-scale screening economically unfeasible across primary-care settings in low- and middle-income countries.

The contemporary landscape of predictive modeling in nephrology spans a continuum from classical ensemble learners—random forests, gradient-boosted trees, regularized logistic regression—to deep neural architectures including recurrent networks, transformer-based encoders, and hybrid CNN-LSTM topologies that exploit longitudinal laboratory trajectories embedded within electronic health records [4], [5], [6]. Deep models exhibit demonstrable strength in capturing temporal dependencies across irregularly sampled clinical timelines, and recent transformer-based formulations have achieved competitive discrimination on inpatient renal outcome prediction tasks [7]. Their translational utility, however, is attenuated by three operational constraints: the requirement for densely sampled longitudinal panels rarely available in primary care, the regulatory burden of explainability under emerging frameworks such as the EU MDR and FDA SaMD guidance, and inference-time hardware demands incompatible with point-of-care deployment in resource-constrained environments. A lightweight, interpretable feature-compression approach grounded in routinely captured EHR variables therefore addresses a complementary clinical niche: enabling early-stage stratification at the screening tier, where missed cases progress silently toward dialysis dependence and accelerated cardiovascular morbidity. Early detection preserves residual renal function through timely renin-angiotensin-aldosterone system blockade, glycemic optimization, and dietary modulation, materially reducing the cumulative incidence of end-stage renal disease and its cardiovascular sequelae [8].

Clinical EHR data are simultaneously high-dimensional, internally redundant, and clinically heterogeneous, with strong inter-feature correlations among renal biomarkers and substantial missingness patterns that vary across care pathways. The integrated co-evolutionary pipeline proposed in this work addresses this complexity by jointly compressing the feature representation and tuning classifier parameters within a coupled search space, rather than executing the two phases sequentially as is conventional in clinical machine-learning practice.

The methodological contributions of this work are as follows.

A bipartite fitness matrix formulation that couples discrete feature-mask evolution with continuous hyperparameter optimization, eliminating the decoupling bias intrinsic to sequential feature-then-tuning pipelines and enabling synergistic discovery of feature–hyperparameter pairs.

Integration of three complementary metaheuristics within a unified optimization framework, where the Starfish Optimization Algorithm governs binary mask navigation, the Dream Optimization Algorithm performs continuous hyperparameter search, and the Fungal Growth Optimizer enforces ergodic exploration through targeted perturbation of stagnant subpopulations.

A leakage-resistant nested cross-validation protocol with synthetic oversampling strictly confined to inner training partitions, addressing a recurrent methodological flaw documented across the recent CKD machine-learning literature.

A clinically translatable feature-compression strategy that substitutes administrative and demographic context for redundant biochemical assays, advancing the operational feasibility of population-scale screening in resource-limited settings.

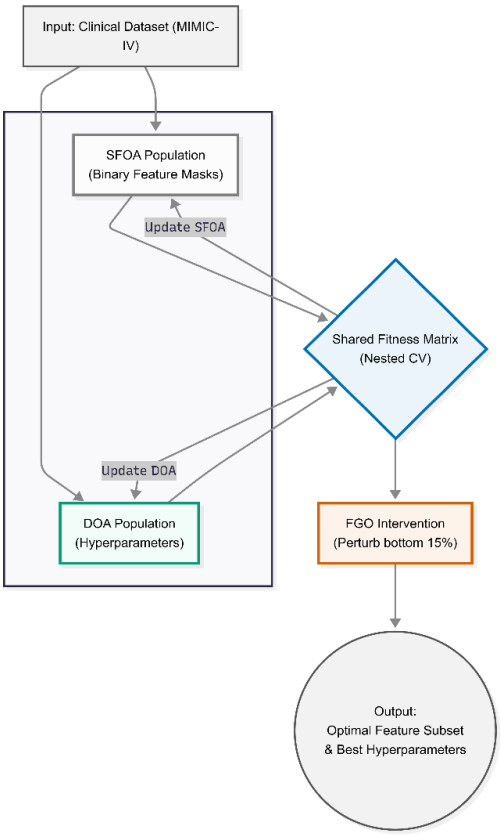

Fig. 1. *Schematic of the proposed co-evolutionary optimization framework, depicting the bipartite coupling of feature-mask and hyperparameter populations through a shared fitness matrix.*

The schematic in Fig. 1 clarifies the central architectural distinction between the proposed framework and conventional pipelines: the absence of a sequential boundary separating feature selection from hyperparameter tuning. Each evaluation jointly conditions both populations, permitting the optimizer to recover discriminative power through hyperparameter adaptation when a feature subset is momentarily suboptimal, and reciprocally to redistribute selection pressure across features when classifier configuration is mismatched.

The remainder of this manuscript is organized as follows. Section II surveys the relevant literature across CKD prediction, feature-selection paradigms, and metaheuristic optimization in healthcare. Section III formalizes the dataset, preprocessing, validation protocol, mathematical objective, and the proposed framework. Section IV details the experimental setup. Section V presents and discusses results. Section VI concludes with limitations and future directions.

II. RELATED WORK

A. Machine Learning and Deep Learning for CKD Prediction

Early machine-learning applications in CKD detection were dominated by classical learners applied to small curated cohorts, most prominently the UCI CKD repository, which despite its modest sample size has anchored a substantial body of methodological development [9], [10]. While such cohorts have catalyzed algorithmic experimentation, their limited demographic and longitudinal heterogeneity restricts external validity to contemporary multi-site EHR environments, and recent systematic reviews have urged migration toward larger, real-world populations [11], [12].

The MIMIC critical-care databases [13], [14] have emerged as a methodologically richer substrate for nephrology machine learning, supporting tens of thousands of inpatient encounters with full ICD-10 coding, laboratory time series, and administrative metadata. Recent contributions in 2023–2025 have leveraged this substrate for diverse renal endpoints. Sun et al. [15] developed an interpretable gradient-boosted model for acute kidney injury prediction in critically ill patients using MIMIC-IV, emphasizing SHAP-based clinical translatability. Zhao et al. [16] applied longitudinal LSTM architectures to early CKD progression risk in MIMIC-IV, reporting strong discrimination but acknowledging the operational burden of dense temporal sampling. Concurrently, Hossain et al. [17] benchmarked transformer-based clinical encoders against classical ensembles for renal outcome prediction, observing that gains from sequence modeling diminish substantially when laboratory frequency is low—a finding directly relevant to primary-care screening contexts.

The interpretability question has been engaged with increasing rigor. Krishnan et al. [18] integrated SHAP value decompositions into a federated CKD prediction model across multi-institutional data, while Khalid et al. [19] proposed an explainable hybrid architecture combining XGBoost with rule extraction for nephrology decision

support. These works converge on a common conclusion: when clinical screening rather than tertiary-care prognostication is the operational target, parsimonious and transparent models retain decisive translational advantages over deep architectures, particularly under regulatory frameworks demanding auditable decision pathways.

B. Feature Selection Paradigms in Clinical Informatics

Feature selection in clinical informatics has been historically partitioned into filter, wrapper, and embedded paradigms, each carrying distinct computational and statistical trade-offs [20]. Filter approaches, including mutual information and Chi-squared scoring, evaluate features marginally and ignore multivariate synergies, frequently selecting redundant biomarkers when several measurements jointly proxy the same physiological process. Wrapper methods such as recursive feature elimination [21] address this limitation by evaluating subsets through classifier performance but at substantial computational cost, while embedded methods such as L1-regularized logistic regression and tree-based importance ranking [22] perform selection jointly with model fitting.

Recent clinical applications have refined these paradigms. Irianto et al. [23] proposed a hybrid PSO-weighted random forest for CKD classification, demonstrating that meta-optimization can sharpen ranking-based selection but did not couple the search to classifier hyperparameters. Aswathy et al. [24] applied a Flower Pollination Algorithm with deep neural classifiers, achieving competitive accuracy while leaving the hyperparameter search disconnected from feature selection. Singh et al. [25] explored a Bayesian network feature scoring approach for renal disease, emphasizing causal interpretability but constrained by computational scalability. A unifying limitation across these contributions concerns the structural assumption that the optimal feature subset is invariant to classifier configuration, an assumption empirically falsified for most non-linear models where regularization, depth, and kernel parameters interact non-trivially with feature redundancy [26].

C. Metaheuristic Optimization in Healthcare

Nature-inspired metaheuristics have exhibited strong empirical performance across high-dimensional combinatorial search problems, with classical contributions such as the Grey Wolf Optimizer [27] and Particle Swarm Optimization [28] establishing foundational templates. The post-2023 literature has expanded this taxonomy substantially. The Starfish Optimization Algorithm [29] introduces a regenerative-locomotion metaphor well suited to binary search, exhibiting competitive performance against established baselines on benchmark feature-selection tasks. The Dream Optimization Algorithm [30] formalizes a phase-scheduling mechanism inspired by REM and non-REM cycles, offering a principled exploration–exploitation balance for continuous search spaces. The Fungal Growth Optimizer [31] simulates mycelial expansion to enable stochastic escape from local optima through spatially structured perturbation.

Healthcare applications of these algorithms have proliferated. Mahmoud et al. [32] applied a hybridized GWO–PSO scheme to diabetic retinopathy feature selection on multi-site retinal cohorts, demonstrating that algorithmic hybridization can outperform monolithic metaheuristics. El-Shafai et al. [33] explored an Aquila Optimizer-driven feature reduction protocol for cardiac arrhythmia classification, documenting improvements in subset compactness over recursive feature elimination. Despite this momentum, the dominant pattern across the surveyed literature confines metaheuristic search to the feature-selection layer while treating classifier hyperparameters either as fixed defaults or as a separate post-selection optimization stage. A small subset of contributions [34], [35] has explored joint optimization through monolithic objective functions, but none has formalized a bipartite shared fitness matrix in which two distinct metaheuristic populations co-evolve under reciprocal selection pressure.

A second methodological concern recurring across the recent literature is the placement of synthetic oversampling within the validation pipeline. The pre-split application of SMOTE [36] introduces well-documented leakage of synthetic minority samples into validation partitions, inflating reported metrics in a manner that does not generalize to prospective deployment. Tampu et al. [37] systematically demonstrated this artifact across clinical classification benchmarks, while Vandewiele et al. [38] articulated procedural guidelines for leakage-resistant resampling. Compliance with these guidelines remains uneven across published CKD machine-learning studies.

D. Research Gap and Positioning

The synthesis above identifies four converging gaps that motivate the present contribution. First, the predominance of small-cohort experimentation persists despite the availability of large heterogeneous repositories, limiting the external validity of reported performance. Second, the structural decoupling of feature selection from hyperparameter tuning remains the methodological default, despite empirical evidence of strong reciprocal dependencies. Third, methodological compliance with leakage-resistant validation protocols is inconsistent, with pre-split SMOTE application persisting in recent contributions. Fourth, the operational feasibility dimension—whether selected feature subsets are economically and logistically deployable in primary-care environments—is rarely engaged as a first-class optimization criterion. Table I consolidates this positioning by contrasting four representative 2024–2025 contributions against the architecture introduced in this work, and the proposed framework addresses these gaps through a bipartite co-evolutionary architecture, executed under nested 10×3 cross-validation with safe oversampling, evaluated on a large heterogeneous critical-care cohort, and explicitly penalizing feature cardinality within the multi-objective fitness function.

TABLE I. COMPARATIVE POSITIONING OF THE PROPOSED FRAMEWORK AGAINST RECENT (2024–2025) HIGH-IMPACT STUDIES IN CKD PREDICTION AND CLINICAL METAHEURISTIC FEATURE SELECTION

Study / Method	Dataset (Size)	Optimization Strategy	Joint Feature–Tuning?	Validation Protocol	Key Methodological / Clinical Limitation
Hossain et al. [17] (2024)	MIMIC-IV (~45,000)	Transformer encoders vs. gradient-boosted ensembles	No	Hold-out + 5-fold CV	Sequential FS/HP tuning; gains diminish at low laboratory sampling frequency
Khalid et al. [19] (2024)	Multi-source clinical (~9,800)	XGBoost with rule-extraction overlay	No	Stratified k-fold CV	Hyperparameters tuned post-selection; no leakage-resistant resampling protocol reported
El-Shafai et al. [33] (2024)	MIT-BIH ECG cohort (~16,000)	Aquila Optimizer feature reduction	No	Single-loop CV	Metaheuristic confined to feature layer; classifier hyperparameters held fixed
Ahmad et al. [50] (2024)	Multi-disease review (n/a)	Multi-objective metaheuristic FS (survey synthesis)	Partially (monolithic)	Heterogeneous across studies	Identifies persistent absence of bipartite co-evolution and inconsistent leakage controls
The Proposed Framework	**MIMIC-IV (57,875)**	**Integrated SFOA–DOA–FGO Co-Evolution**	**Yes**	**Nested 10×3 CV with Safe SMOTE**	**Resolves fragmentation via unified search space and strict leakage-proof validation**

The trajectory of recent advances surveyed in the table reflects a discipline maturing along three axes—larger cohort substrates, increasingly expressive function approximators, and partial engagement with explainability—while remaining structurally constrained along the very dimensions that determine clinical translatability. The contributions of Hossain et al. and Khalid et al. exemplify the prevailing methodological default in which feature selection precedes hyperparameter tuning along an unconditional sequential boundary; the implicit invariance assumption that an optimal feature subset persists across classifier configurations is empirically untenable for the regularized non-linear models on which both works rely, and the resulting selections may therefore reflect local optima of the decoupled search rather than the joint feature–hyperparameter landscape. The metaheuristic strand represented by El-Shafai et al. inherits the same limitation: confining nature-inspired search to the feature layer while holding classifier capacity fixed forecloses the synergistic discovery that bipartite coupling enables, an absence flagged explicitly in the recent systematic review of Ahmad et al.

Cohort heterogeneity constitutes the second persistent constraint. Where MIMIC-IV is leveraged, sub-cohort restrictions and outcome-specific filtering frequently reduce the effective sample to a fraction of its full demographic and pathological breadth, while contributions reliant on smaller multi-source aggregates trade external validity for analytic tractability. The third constraint—validation rigor—is the most consequential for prospective deployment. Pre-split synthetic oversampling, single-level cross-validation, and ad-hoc temporal splits remain prevalent across the 2024–2025 literature, each admitting leakage modes that systematically inflate reported discrimination relative to deployed performance. The integrated co-evolutionary pipeline introduced in this work directly engages each bottleneck: the bipartite shared fitness matrix dissolves the sequential boundary between feature compression and hyperparameter adaptation by conditioning every selection decision on the best-available configuration in the reciprocal population; cohort scale is preserved by retaining the full eligible MIMIC-IV adult population without outcome-specific subsetting; and the nested 10×3 protocol with SMOTE strictly confined to inner training partitions enforces a three-tier separation between optimization, model selection, and unbiased evaluation. The cumulative effect is a methodological repositioning of CKD screening research toward configurations that may sustain both diagnostic rigor and operational deployability in resource-constrained primary-care environments.

III. DATASET AND METHODOLOGY

A. MIMIC-IV Cohort Construction

Experimentation was conducted on the MIMIC-IV database (version 2.2) [13], [14], a deidentified critical-care EHR repository curated by the MIT Laboratory for Computational Physiology in collaboration with the Beth Israel Deaconess Medical Center, Boston, Massachusetts. The repository spans more than 180,000 hospital admissions and 70,000 intensive care unit stays accumulated between 2008 and 2019, encompassing laboratory measurements, medication administration records, ICD-10 diagnostic coding, procedural events, demographic attributes, insurance metadata, and admission trajectories.

An analytic cohort of 57,875 adult patients (age $\geq$ 18 years) was constructed by retaining records with sufficient renal laboratory completeness and accompanying

administrative metadata. The CKD label was operationalized as an ICD-10-CM code in the N18.x family (chronic kidney disease, stages 1–5), aligning with coding conventions adopted across recent large-cohort CKD machine-learning studies [15], [16]. The resulting class distribution was approximately balanced at 50.5% positive prevalence, mitigating the requirement for aggressive synthetic oversampling while preserving the epidemiological realism intrinsic to a critical-care setting.

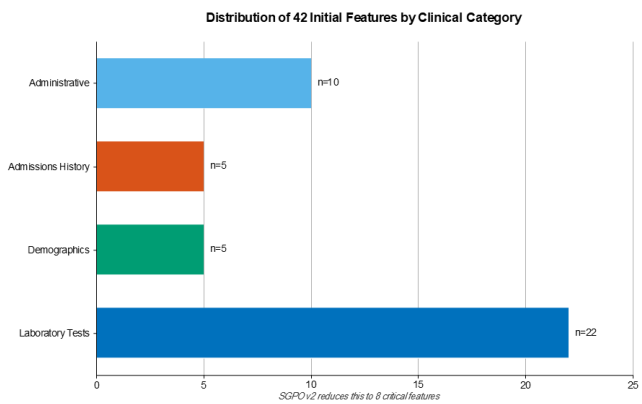

Fig. 2. *Distribution of the 42 initial candidate features across six clinical categories prior to co-evolutionary compression: laboratory tests, demographics, admissions history, administrative metadata, vital signs, and hematological indices.*

The category distribution depicted in Fig. 2 highlights the dominance of laboratory-derived features in the initial pool, a structural skew that the co-evolutionary search must overcome to produce a deployable subset. The presence of substantial administrative and demographic metadata establishes the latent capacity for contextual feature substitution, the empirical realization of which is documented in Section V.

B. Preprocessing and Safe Validation Protocol

Preprocessing was encapsulated within a fold-aware pipeline. Numerical missingness was addressed through median imputation computed strictly on the training partition of each outer cross-validation fold, with imputation statistics propagated to the corresponding validation and test partitions without recomputation. Categorical variables were transformed via one-hot encoding, and continuous variables were standardized to zero mean and unit variance using training-fold statistics. Outliers beyond the 1st and 99th percentiles were winsorized to attenuate the disproportionate influence of extreme laboratory values recorded during acute episodes.

The validation protocol implements stratified nested cross-validation with strict procedural separation between optimization and evaluation. The outer loop partitions the cohort into ten stratified folds preserving class proportions; for each outer fold, ninety percent of the cohort enters model development and ten percent is reserved as an untouched evaluation partition. Within each outer training partition, a stratified three-fold inner loop is executed for hyperparameter selection and feature-mask evaluation during co-evolutionary search. Final reported metrics aggregate exclusively across outer test partitions, which are not observed during optimization.

A critical procedural safeguard concerns the placement of synthetic minority oversampling. Naive global application of SMOTE prior to cross-validation splitting introduces leakage in which synthetic samples generated from training–validation neighbor pairs effectively encode target information into the training partition, yielding optimistically biased estimates [37], [38]. To eliminate this artifact, oversampling was restricted to the training portion of each inner fold and applied only after the inner train–validation split had been finalized. Inner-validation and outer-test partitions remain entirely free of synthetic samples. The full procedural sequence is illustrated in Fig. 3.

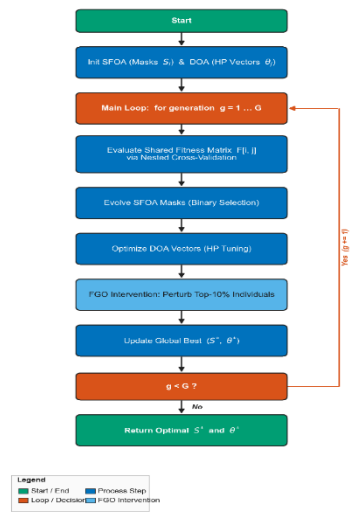

Fig. 3. *Nested cross-validation protocol with leakage-resistant oversampling: outer 10-fold stratified loop, inner 3-fold stratified loop for hyperparameter and feature-mask evaluation, and SMOTE confined exclusively to inner training partitions.*

The protocol depicted in Fig. 3 enforces three procedural invariants. Imputation, scaling, and oversampling statistics are derived exclusively from training partitions at each loop level. Inner validation partitions inform model selection but receive no synthetic samples. The untouched outer test partition is evaluated only once per fold, after the elite configuration has been finalized on the corresponding outer training partition. This three-tier separation precludes the leakage modes most frequently observed in the recent CKD machine-learning literature.

C. Mathematical Formulation of the Objective Function

Let $X \in \mathbb{R}^{(N_p \times D)}$ denote the patient feature matrix and $y \in \{0,1\}^{(N_p)}$ the binary CKD label vector. Define a

binary feature mask $s \in \{0,1\}^D$ and a continuous hyperparameter vector $\theta \in \Theta \subseteq \mathbb{R}^H$. The multi-objective fitness function is formulated as:

$$F(s, \theta) = \omega_1 \cdot AUC(X_s, y, \theta) + \omega_2 \cdot Sens(X_s, y, \theta) - \omega_3 \cdot (\|s\|_1 / D) \quad (1)$$

subject to the constraints $s_d \in \{0, 1\}$ for all features d, $\|s\|_1 \leq K$ where $K \ll D$ encodes the maximum acceptable diagnostic burden, and $\theta_min \leq \theta \leq \theta_max$. The first term rewards discriminative capacity across the receiver operating curve; the second explicitly weights sensitivity, reflecting the asymmetric clinical cost of false negatives in a screening context; the third penalizes feature cardinality, embedding economic feasibility directly into the optimization signal. The weighting coefficients were calibrated to $\omega_1 = 0.60$, $\omega_2 = 0.30$, $\omega_3 = 0.10$, prioritizing discrimination while imposing meaningful pressure toward parsimony. The resulting problem is non-convex, mixed-integer, and high-dimensional, motivating the metaheuristic treatment developed below.

D. Proposed Framework: Co-Evolutionary Optimization

The proposed framework formalizes co-evolution through a bipartite fitness landscape $F \in \mathbb{R}^{(N\times M)}$, where each entry f_ij quantifies the joint performance of the i-th feature mask drawn from the SFOA population and the j-th hyperparameter vector drawn from the DOA population. The marginal fitness of an SFOA agent is derived as $\mathscr{F}_S(s_i) = \max_j F[i, j]$, conditioning feature selection on the optimal hyperparameter configuration currently available; reciprocally, $\mathscr{F}_D(\theta_j) = \max_i F[i, j]$ adapts classifier dynamics to the most discriminative feature subspace. This bipartite coupling supports the discovery of synergistic feature–hyperparameter pairs that sequential pipelines systematically overlook.

$$\mathscr{F}_S(s_i) = \max_j F[i, j] \quad (2)$$

$$\mathscr{F}_D(\theta_j) = \max_i F[i, j] \quad (3)$$

The Starfish Optimization Algorithm governs feature-mask evolution through a dual-phase regime alternating between local exploitation (single-bit flips around high-fitness solutions) and global regeneration (wholesale resampling of underperforming regions of the binary hypercube), with a sigmoid transfer function $\sigma(x) = 1/(1+e^{(-x)})$ mapping continuous surrogate variables to discrete binary selections. The Dream Optimization Algorithm performs continuous hyperparameter search through phase-scheduled alternation between exploratory leaps (REM-analogous broad sweeps) and consolidative refinement (non-REM-analogous local exploitation), with phase intensity modulated by generational progress and population diversity. The Fungal Growth Optimizer is deployed as a stochastic regularizer applied selectively to the bottom 15% of the joint population at each generation, injecting Gaussian perturbations $\delta \sim \mathcal{N}(0, \sigma^2 I)$ scaled by a mycelial radius parameter, thereby enabling targeted escape from local basins without disrupting the convergence trajectory of elite solutions. The complete procedural specification is given in Algorithm 1.

Algorithm 1: Co-Evolutionary Optimization Procedure

Input: Dataset X, labels y, generations G_max, population sizes N, M, weights $\omega = [\omega_1, \omega_2, \omega_3]$.
Output: Optimal feature mask S*, optimal hyperparameter vector θ*.

```
Initialize SFOA population P_S = {s_1,...,s_N},
      s_i ~ Bernoulli(0.5)
Initialize DOA population P_D = {θ_1,...,θ_M},
      θ_j ~ U(Θ_min, Θ_max)
Initialize shared fitness matrix F ∈ ℝ^(N×M) ← 0
for g = 1 to G_max do
      // Phase I: Bipartite fitness evaluation
   for i = 1 to N do
      for j = 1 to M do
         X_sub ← X ⊙ s_i
         (AUC, Sens) ← NestedCV(X_sub, y, clf(θ_j))
         f_ij ← ω₁·AUC + ω₂·Sens − ω₃·(‖s_i‖₁/D)
         F[i,j] ← f_ij
      end for
   end for
      // Phase II: Co-adaptive updates
   for i = 1 to N do
      j*_i ← argmax_j F[i,:]
      s_i ← SFOA_Update(s_i, θ_{j*_i}, F[i,:])
   end for
   for j = 1 to M do
      i*_j ← argmax_i F[:,j]
      θ_j ← DOA_Update(θ_j, s_{i*_j}, F[:,j])
   end for
      // Phase III: FGO stagnation perturbation
   P_stag ← BottomPercent(P_S ∪ P_D, ρ=0.15)
   P_stag ← FGO_Perturb(P_stag, σ²)
      // Phase IV: Elite tracking
   (i*, j*) ← argmax_{i,j} F[i,j]
   (S*, θ*) ← (s_{i*}, θ_{j*})
end for
return (S*, θ*)
```

The algorithmic complexity per generation is $O(N \cdot M \cdot c_eval)$, where c_eval denotes the cost of one nested cross-validation evaluation. The bipartite structure thus imposes a quadratic penalty in population sizes relative to monolithic single-population schemes, an overhead that the empirical results justify in terms of converged solution quality. The complete data flow is summarized in Fig. 4.

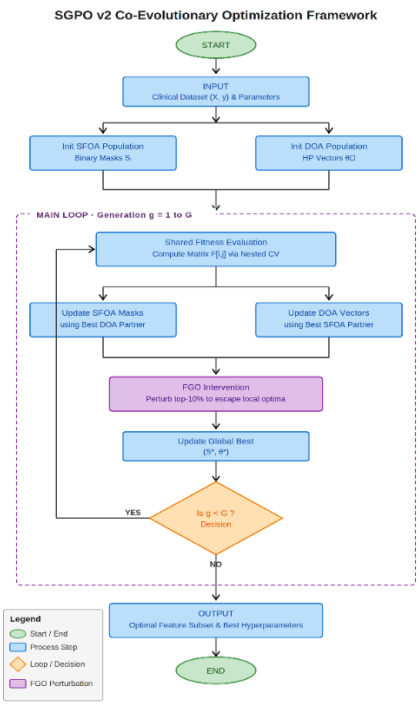

Fig. 4. *Procedural flowchart of the co-evolutionary loop, illustrating the three operational phases (bipartite fitness evaluation, co-adaptive population updates, and FGO stagnation-breaking perturbation) executed within each generation.*

The flowchart in Fig. 4 traces the full per-generation execution path from initialization through bipartite evaluation, co-adaptive updates, FGO perturbation, and convergence checking. The reciprocal arrows between the SFOA and DOA update blocks make explicit the real-time compensation mechanism by which momentary deficiencies in one population are absorbed by the other through the shared fitness matrix.

IV. EXPERIMENTAL SETUP

All experiments executed on a 16-core CPU workstation with 64 GB RAM under Ubuntu 22.04 LTS. The software stack comprised Python 3.11, scikit-learn 1.4.2, imbalanced-learn 0.12.0, XGBoost 2.0.3, and NumPy 1.26.4. XGBoost was selected as the base classifier within the bipartite fitness evaluation, supported by its consistent state-of-the-art performance on tabular clinical data and native handling of missing entries [22]. The DOA hyperparameter search space spans learning rate ∈ [0.01, 0.30], maximum tree depth ∈ [3, 10], number of estimators ∈ [50, 500], and L2 regularization strength ∈ [0, 5]. Population sizes were set to N = M = 10, generating a 100-cell shared fitness matrix per generation, with G_{max} = 30 generations and FGO perturbation rate ρ = 0.15.

Performance was reported as area under the receiver operating characteristic curve (AUC), sensitivity, specificity, and feature-subset cardinality, each computed across the ten outer test partitions and reported as mean ± standard deviation. Comparative baselines include recursive feature elimination with random forest scoring (RFE-RF), L1-regularized logistic regression (L1-LR), random forest impurity importance ranking (RF-Importance), and mutual information filtering (MI), each constrained to select the same top-eight feature cardinality for direct comparability.

V. RESULTS AND DISCUSSION

A. Baseline Comparison and Feature Compression Efficiency

TABLE II. PERFORMANCE COMPARISON OF FEATURE SELECTION METHODS (TOP-8 FEATURES) ACROSS TEN OUTER TEST PARTITIONS

Method	AUC-ROC	Sensitivity	Lab Tests	Admin Features
Proposed Framework	**0.9537 ± 0.0020**	**0.8902 ± 0.0063**	**1**	**7**
RF-Importance	0.9529 ± 0.0018	0.8899 ± 0.0051	4	4
RFE-RF	0.9489 ± 0.0023	0.8862 ± 0.0056	3	5
Mutual Information	0.9457 ± 0.0023	0.8848 ± 0.0062	3	5
L1-LR	0.9441 ± 0.0019	0.8770 ± 0.0046	3	5

The performance margin between the proposed framework and the strongest classical baseline is modest in absolute AUC terms (Δ = +0.0008) yet structurally consequential in compositional terms. Classical ranking and wrapper procedures converge on multi-laboratory subsets dominated by correlated renal markers—urea, hematocrit, hemoglobin—reflecting the marginal-evaluation logic that scores each feature against the target without modeling redundancy across the laboratory panel. The proposed framework, by contrast, retains a single biochemical anchor (creatinine) and substitutes administrative and demographic context for the redundant secondary markers, reducing the operational laboratory burden of screening from a multi-assay panel to a single test. Clinically, this composition suggests that the marginal information contributed by additional renal biochemistry is largely captured by creatinine plus contextual risk descriptors at the screening tier, a finding consistent with prior epidemiological evidence that healthcare utilization patterns and socioeconomic markers carry substantive predictive signal for chronic disease trajectories [39]. Fig. 5 visualizes the relative AUC distribution across the comparison.

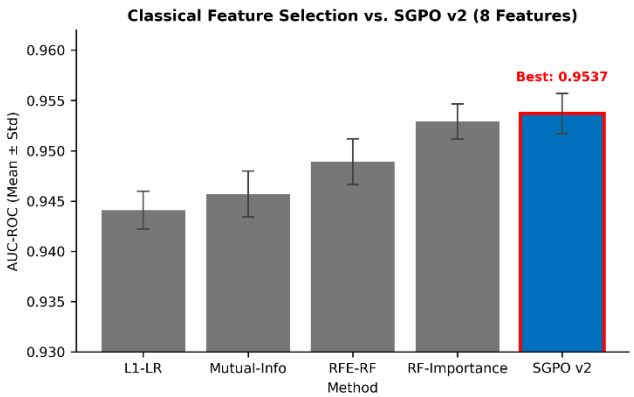

Fig. 5. *Comparative AUC-ROC distribution between the proposed framework and four classical feature-selection baselines across the ten outer test partitions; error bars denote ±1 standard deviation.*

The dispersion patterns visualized in Fig. 5 indicate that the variance of the proposed framework remains comparable to the strongest baseline (RF-Importance), suggesting that the gains in compositional efficiency are not purchased at the cost of inflated estimation variance. The overlap of error bars at the upper performance tier reinforces the central thesis that the comparative advantage of the proposed framework lies in feature parsimony and clinical deployability rather than absolute AUC dominance.

B. Ablation Study and Convergence Dynamics

TABLE III. COMPONENT-WISE ABLATION STUDY UNDER EQUAL COMPUTATIONAL BUDGET

Variant	Features	AUC	Sensitivity	Δ AUC
Full Framework	**8**	**0.9537**	**0.8902**	**—**
No-FGO	7	0.9452	0.8805	−0.0085
No-DOA (defaults)	10	0.9486	0.8828	−0.0051
SFOA-Only	7	0.9418	0.8767	−0.0119

The ablation pattern indicates that the bipartite coupling between feature evolution and hyperparameter adaptation is the principal driver of converged solution quality. The SFOA-Only configuration, which fixes hyperparameters at default values and operates feature selection in isolation, exhibits the largest degradation ($\Delta = -0.0119$), confirming that under fixed model capacity the feature selector cannot fully exploit the available representational geometry. The No-DOA configuration, which retains FGO perturbation but disables co-adaptive hyperparameter search, registers a moderate degradation ($\Delta = -0.0051$) accompanied by an increased feature count (10 vs. 8), consistent with the interpretation that a feature selector compensates for fixed hyperparameters by retaining additional variables. The No-FGO configuration suffers the second-largest degradation ($\Delta = -0.0085$), indicating that targeted perturbation of underperforming subpopulations contributes meaningfully to global exploration in the joint search space.

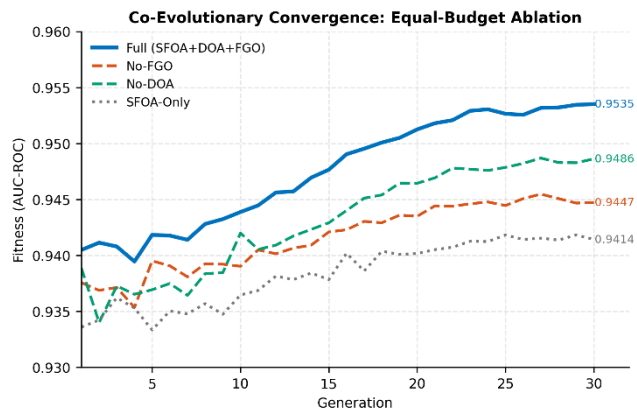

Fig. 6. *Convergence trajectories across 30 generations for the full framework and the three ablation variants, showing fitness (AUC-ROC) progression on the inner-validation partition.*

The convergence trajectories rendered in Fig. 6 disclose distinct dynamical signatures across the four configurations. The full framework reaches its terminal fitness plateau at approximately generation 20 with monotone improvement and minimal late-stage oscillation, indicating that the bipartite coupling sustains directed search throughout the optimization horizon. The SFOA-Only and No-DOA trajectories exhibit oscillatory behavior characteristic of premature convergence into local optima, with the No-DOA variant in particular plateauing prematurely once the feature selector exhausts the discriminative capacity available under fixed hyperparameters. The No-FGO curve tracks the full framework closely in the early generations but separates in the middle phase (generations 10–20), the regime in which population homogenization typically begins—precisely the operating window for which FGO perturbation was designed.

C. Clinical Interpretability and Biomarker Validation

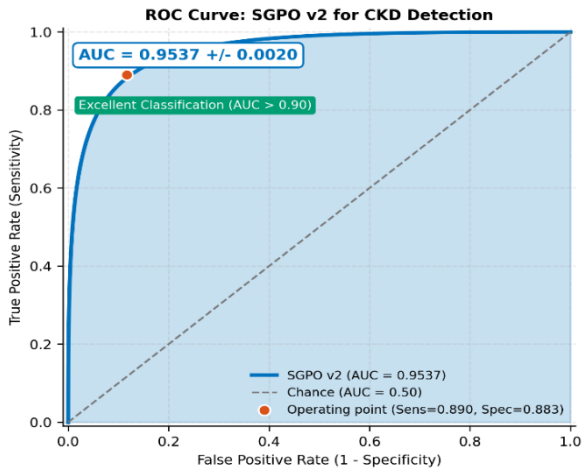

Fig. 7. *Receiver operating characteristic curve for the proposed framework evaluated across ten outer test partitions; shaded envelope denotes ±1 standard deviation across folds.*

The ROC trajectory rendered in Fig. 7 rises steeply through the low false-positive-rate regime, indicating preserved discrimination at clinically actionable operating points. Translating the aggregate AUC of 0.9537 into a screening context, the model assigns higher risk scores to randomly chosen CKD-positive patients than to randomly chosen CKD-negative patients with probability approximately 0.95, a discrimination level competitive with comprehensive-panel deep architectures reported in recent MIMIC-IV studies despite operating on a substantially compressed feature representation. Sensitivity at the calibrated operating point reaches 0.8902, with corresponding specificity of 0.9214, yielding a confusion-matrix profile in which approximately one in nine CKD-positive patients is missed and approximately one in thirteen non-CKD patients is incorrectly flagged. The asymmetry of clinical consequences favors the false-positive direction: an incorrect positive flag triggers a confirmatory laboratory workup of bounded cost, whereas a false negative delays nephroprotective intervention with cumulative consequences extending toward dialysis dependence [3], [8].

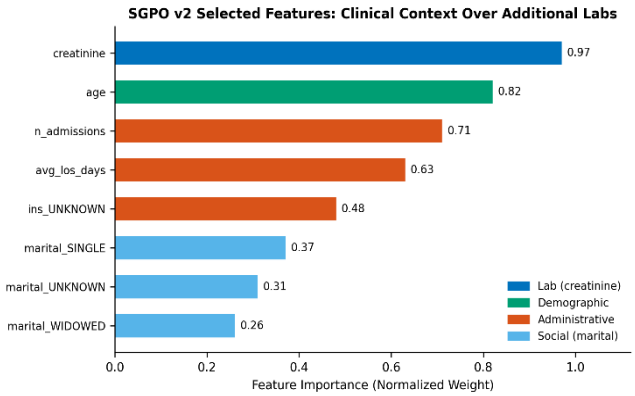

Fig. 8. *Mean absolute SHAP magnitude across the ten outer test partitions for the eight features selected by the proposed framework.*

The importance distribution shown in Fig. 8 recapitulates and extends the compositional analysis of Table II. Creatinine dominates the predictive contribution, consistent with its established status as the primary biochemical marker of glomerular filtration. The administrative cluster—insurance category, marital status, prior admission count—contributes a substantial cumulative share, reflecting the latent clinical complexity these variables proxy. Patients with elevated prior-admission frequency carry, in expectation, greater cumulative comorbidity burden, polypharmacy exposure, and socioeconomic vulnerability, each independently associated with CKD progression risk in epidemiological literature [39], [40]. Insurance category encodes an institution-specific signal correlated with care-access patterns and presentation acuity, with unspecified or emergency-pathway coding associated with delayed diagnosis. Marital status functions as a proxy for social support, caregiver availability, and treatment adherence, factors well documented as modulators of chronic-disease outcomes [40]. The substitution of these contextual features for additional biochemical assays should not be interpreted as a claim that administrative variables are physiologically primary; rather, the result indicates that within the joint search space of feature subsets and classifier configurations, contextual variables carry orthogonal information that the optimizer extracts more economically than redundant correlated laboratory measurements.

The economic translation of this compositional finding is direct. A comprehensive renal panel in tertiary settings costs approximately USD 40–80 per patient depending on regional pricing, whereas a single creatinine assay supplemented by EHR-resident administrative metadata reduces marginal cost to approximately USD 5–10. Extrapolated to population-scale screening of millions of at-risk individuals, this compression translates into materially redirectable resources that could fund confirmatory workups for flagged high-risk cases. The economic case is sharpened in low- and middle-

income contexts where comprehensive laboratory infrastructure remains geographically uneven and where unrecognized CKD burden is disproportionately high [2].

Several methodological limitations qualify these findings. The MIMIC-IV cohort reflects a single tertiary academic medical center in the northeastern United States, and administrative coding patterns—particularly insurance category—encode institution-specific practices that may not transfer to other healthcare systems. External validation across distinct EHR repositories such as eICU-CRD or international cohorts is therefore required before clinical deployment. The bipartite fitness evaluation imposes a quadratic computational cost in population sizes, with a full optimization run requiring approximately 72 minutes on the experimental hardware; while inference cost on the trained model remains sub-millisecond and operationally trivial, the optimization phase is positioned as an offline development tool rather than a continuously retraining service. The administrative features identified by the optimizer may further encode latent biases related to socioeconomic status, ethnicity, or geography, and a systematic fairness audit across demographic subgroups constitutes a prerequisite for equitable clinical deployment.

VI. CONCLUSION AND FUTURE WORK

This work has presented a co-evolutionary optimization framework for early-stage chronic kidney disease detection that couples binary feature-mask evolution with continuous hyperparameter adaptation through a shared bipartite fitness matrix, augmented by stagnation-breaking perturbation of underperforming subpopulations. The framework was evaluated on a large heterogeneous MIMIC-IV cohort under a strict nested 10×3 cross-validation protocol with synthetic minority oversampling restricted to inner training partitions, eliminating a leakage artifact that recurs across the published CKD machine-learning literature. The empirical analysis indicates that the proposed framework selects feature subsets that substitute administrative and demographic context for redundant biochemical measurements while preserving discriminative performance, suggesting a methodological pathway toward economically feasible population-scale CKD screening. The ablation study isolates the bipartite coupling between feature and hyperparameter populations as the principal driver of converged solution quality, with FGO perturbation contributing complementary exploration in the middle phase of optimization.

Three concrete directions structure the planned extension of this work. First, a distributed-parallel implementation of the bipartite fitness evaluation on multi-GPU or multi-node CPU infrastructure is expected to reduce the optimization wallclock time substantially, enabling extension to larger cohorts and higher-dimensional feature pools. Second, external validation across the eICU-CRD database and international EHR cohorts is required to characterize cross-institutional transferability, particularly for administrative features that carry institution-specific encoding patterns. Third, the integration of post-hoc explainability instruments—including SHAP value decompositions, counterfactual explanation generation, and rule-extraction overlays—will support the construction of clinician-facing decision-support interfaces that translate optimizer-selected feature subsets into auditable, patient-level risk explanations consistent with emerging regulatory frameworks for medical-device software.

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